
Smart Crop Advisory System for Small and Marginal Farmers

1. Executive Summary

- Agriculture is the backbone of India's economy.
- 80% of Indian farmers are **small and marginal**, struggling with low productivity, unpredictable weather, and poor market access.
- Existing advisory services are **generic, inaccessible, or too costly**.
- **Smart Crop Advisory System (SCAS)** proposes a **digital-first, AI-driven, low-cost, and multilingual platform** that delivers:
 - Soil & crop recommendations.
 - Pest/disease detection.
 - Weather-based alerts.
 - Market price updates.



2. Problem Statement

Key Issues:

- Wrong crop selection → poor yields.
- 30–40% losses from pests/diseases.
- Erratic rainfall and climate shocks.

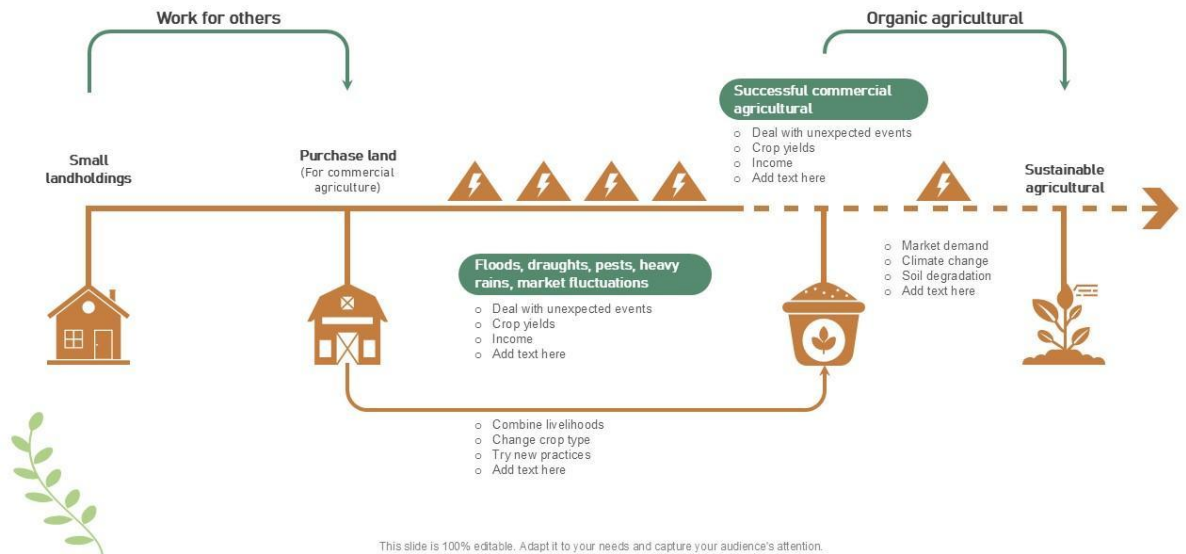
- Exploitation due to lack of price info.
- Digital divide (smartphone vs feature phone).



3. Objectives

- Provide **personalized, real-time advisory**.
- Support **feature phone users** with SMS/voice.
- Reduce losses, improve yields, and increase income.
- Promote **sustainable farming** (balanced fertilizer, organic pest control)

Roadmap to more sustainable agriculture



4. Proposed Solution – Smart Crop Advisory System

Core Modules:

1. **Crop Recommendation** (based on soil, location, season).
2. **Pest Detection** (AI photo recognition).
3. **Weather Alerts** (SMS/voice in local language).
4. **Market Price Info** (real-time mandi prices).

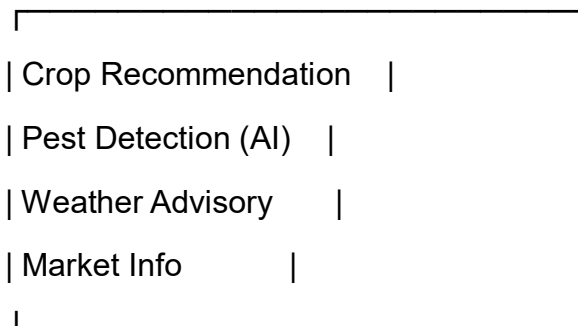


System Diagram

Farmer Input (Soil / Photo / Location)



Backend Engine



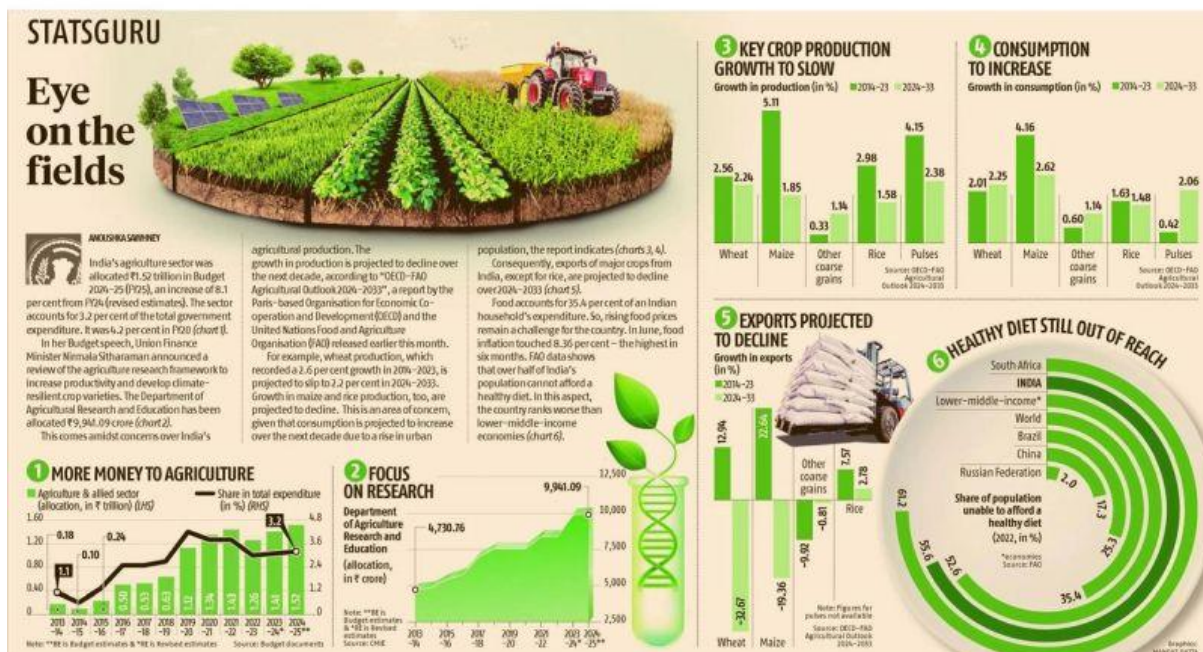
Output: Mobile App / SMS / IVR



5. Technology Stack

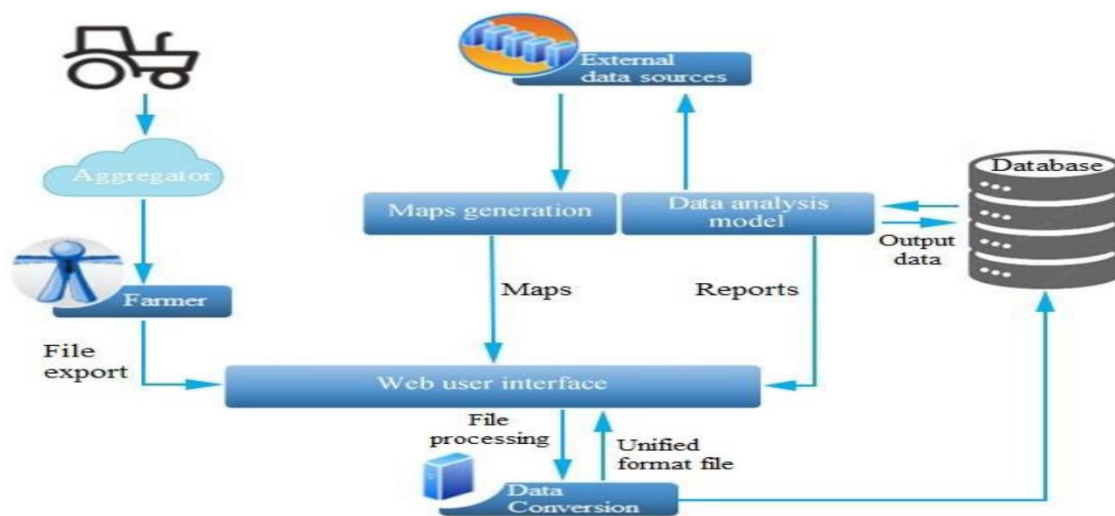
Layer	Tech Used
Frontend	React Native / Flutter
Backend	Node.js, Firebase/AWS
Database	Firebase DB, PostgreSQL
AI/ML	TensorFlow Lite (CNN for pest)
APIs	OpenWeather, FAO/ICAR datasets
IoT (optional)	Soil moisture & pH sensors

BUSINESS STANDARD,MUMBAI 29 JULY 2024



6. System Workflow (11)

1. **Register farmer** (local language, simple form).
2. **Input soil type/location** → suggests crops.
3. **Upload crop photo** → AI detects pest/disease.
4. **Weather API** → irrigation & fertilizer alerts.
5. **Mandi API** → price updates via SMS.



7. Demo Flow

Scenario:

- Soil: Red Sandy Loam.
- Recommendation: Tomato, Groundnut, Cotton.
- Farmer uploads tomato leaf → AI detects blight.
- Cure: Neem oil + fungicide.
- Rain forecast → SMS: “Skip irrigation today.”
- Market update → “Sell groundnut after 3 days.”

Plant Disease Detection App (v.2)

Watch Video to know more



8. Creative Diagrams

a) Data Flow Diagram



b) Architecture Layers

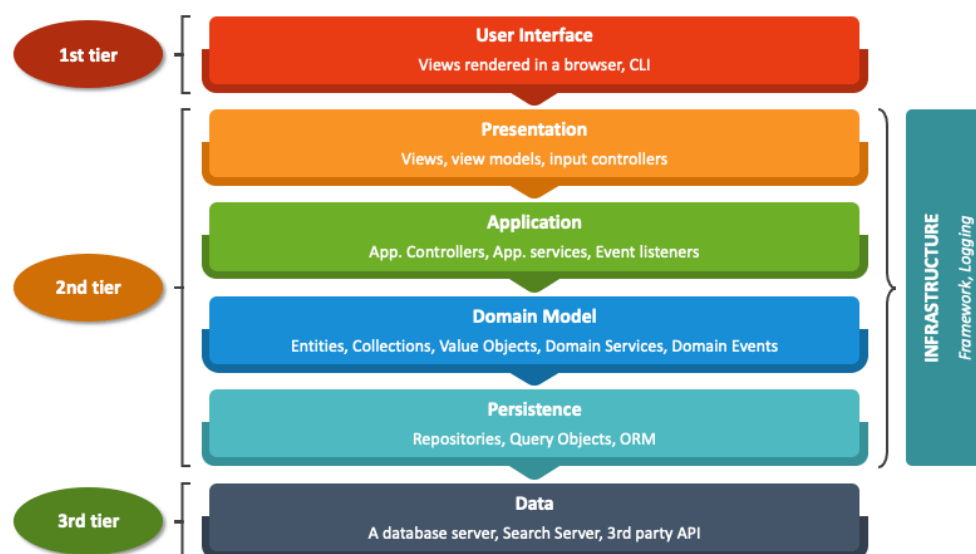
[User Layer] → App / SMS / IVR

[Service Layer] → Crop advisory, pest detection

[Data Layer] → Weather + soil DB + market API

[Infrastructure] → Cloud, AI models, IoT devices

LAYERED ARCHITECTURE



9. Uniqueness (16)

- **Localized:** Supports multiple Indian languages.
- **Inclusive:** Feature phone + smartphone.
- **Low-cost:** Uses open datasets and lightweight AI.
- **Actionable:** Clear “do this today” advice.



10. Impact

Metric	Current With SCAS	
Yield/acre	Baseline +20%	
Crop loss	40%	-20%
Input cost	High	-15%

Metric Current With SCAS

Market awareness Low High

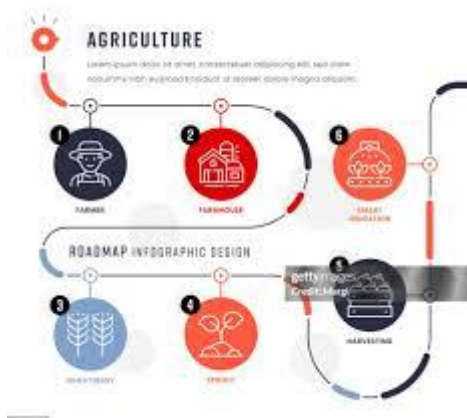


11. Implementation Roadmap

Phase 1 (0–3 months): Hackathon MVP.

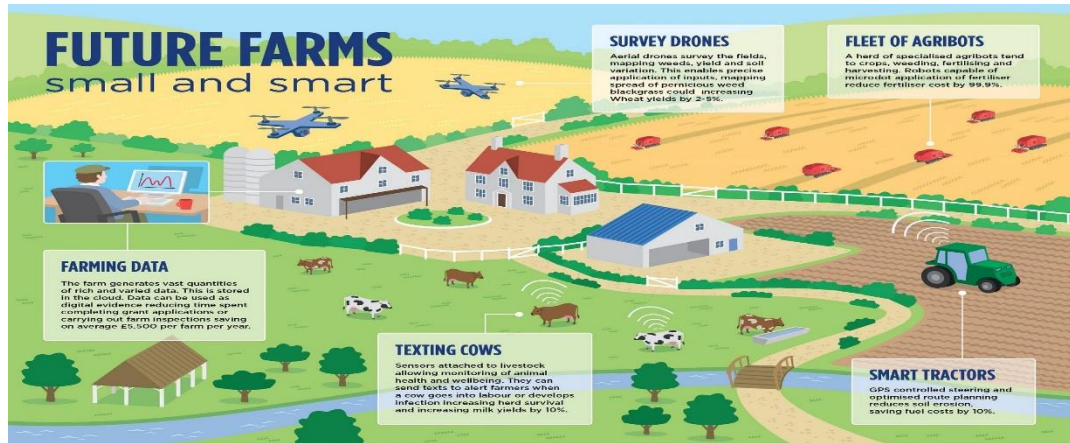
Phase 2 (3–12 months): Pilot with 500 farmers.

Phase 3 (12+ months): Scale-up with Govt & NGOs.



12. Future Scope

- Drones for crop monitoring.
- IoT soil sensors for precision farming.
- Blockchain for produce traceability.
- AI-driven market forecasting.



13. Conclusion

The **Smart Crop Advisory System** will be a **game-changer** for small and marginal farmers by bridging the digital divide. It ensures:

- Better yields.
- Reduced risks.
- Higher incomes.
- Sustainable farming practices.



Thank
you!

